

Number Theory and Proof Methods

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4.1 Introduction

4.2 Rational Numbers

 **4.3 Divisibility**

4.4 Quotient-Remainder Theorem



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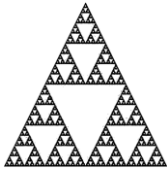


<http://jarrar-courses.blogspot.com/2014/03/discrete-mathematics-course.html>

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Acknowledgement:

This lecture is based on, but not limited to, chapter 3 in “Discrete Mathematics with Applications by Susanna S. Epp (3rd Edition)”.



Number Theory

4.3 Divisibility

In this lecture:

-  ☐ Part 1: **What is Divisibility;**
- ☐ Part 2: Proving Properties of Divisibility;
- ☐ Part 3: The Unique Factorization Theorem

What is Divisibility?

• Definition

If n and d are integers and $d \neq 0$ then

n is **divisible by** d if, and only if, n equals d times some integer.

Instead of “ n is divisible by d ,” we can say that

n is a **multiple of** d , or

d is a **factor of** n , or

d is a **divisor of** n , or

d **divides** n .

The notation $d \mid n$ is read “ d divides n .” Symbolically, if n and d are integers and $d \neq 0$:

$$d \mid n \iff \exists \text{ an integer } k \text{ such that } n = dk.$$

Examples

✓ Is 21 divisible by 3?

✓ Does 5 divide 40?

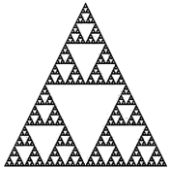
✓ Does $7 \mid 42$?

✓ Is 32 a multiple of -16 ?

✓ Is 6 a factor of 54?

✓ Is 7 a factor of -7 ?

✓ If k is any integer, does k divide **0**?



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Positive Divisor of a Positive Integer

Theorem 4.3.1 A Positive Divisor of a Positive Integer

For all integers a and b , if a and b are positive and a divides b , then $a \leq b$.

Proof:

$$b = a.k$$

Thus

$$1 \leq k$$

$$a.1 \leq k . a$$

multiply both sides with a .

Thus

$$a \leq k . a = b$$

Thus

$$a \leq b$$

Divisibility of Algebraic Expressions

If a and b are integers, is $3a + 3b$ divisible by 3?

$3a + 3b = 3(a + b)$ and $a + b$ is an integer because it is a sum of two integers.

If k and m are integers, is $10km$ divisible by 5?

$10km = 5 \cdot (2km)$ and $2km$ is an integer because it is a product of three integers.

Not divisible

For all integers n and d , $d \nmid n \Leftrightarrow \frac{n}{d}$ is not an integer.

Prime Numbers and Divisibility

An alternative way to define a prime number is to say that:

an integer $n > 1$ is prime if, and only if, its only positive integer divisors are 1 and itself.

Transitivity of Divisibility

Theorem 4.3.3 Transitivity of Divisibility

For all integers a , b , and c , if a divides b and b divides c , then a divides c .

Proof:

Starting Point: Suppose a , b , and c are particular but arbitrarily chosen integers such that $a \mid b$ and $b \mid c$.

We need to show: $a \mid c$.

since $a \mid b$, $b = ar$ for some integer r .

And since $b \mid c$, $c = bs$ for some integer s .

Hence, $c = bs = (ar)s$

But $(ar)s = a(rs)$ by the associative law

Hence $c = a(rs)$.

As rs is an integer, then $a \mid c$.

Divisibility by a Prime

Theorem 4.3.4 Divisibility by a Prime

Any integer $n > 1$ is divisible by a prime number.

Theorem: Every integer $n > 1$ is divisible by at least one prime number.

Proof (By Contradiction):

Assume the Negation: Suppose there exists at least one integer $n > 1$ that is not divisible by any prime number.

Define a Set: Let S be the set of all integers greater than 1 that have no prime divisors. Based on our assumption, S is non-empty.

Apply Well-Ordering Principle: Since S is a non-empty set of positive integers, it must have a least element, call it m . This means m is the smallest integer greater than 1 that is not divisible by any prime.

Analyze m : * Is m prime? No. If m were prime, it would be divisible by itself (a prime), which contradicts m being in S .

Therefore, m must be composite.

Decompose m : Since m is composite, there exist integers a and b such that $m = a * b$, where $1 < a < m$.

Finding the Contradiction: * Because $a < m$ and m is the least element in S , a cannot be in S .

Since a is not in S and $a > 1$, a must have a prime divisor, let's call it p .

So, $p \mid a$. And we know that $a \mid m$.

By the Transitivity of divisibility: If $p \mid a$ and $a \mid m$, then $p \mid m$.

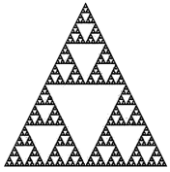
Conclusion: We found a prime p that divides m . This contradicts our initial assumption that m has no prime divisors. Thus, the original theorem must be true.

Counterexamples and Divisibility

Checking a Proposed Divisibility Property

Is it true or false that for
all integers a and b , if $a \mid b$ and $b \mid a$ then $a = b$?

Counterexample: Let $a = 2$ and $b = -2$. Then
 $a \mid b$ since $2 \mid (-2)$ and $b \mid a$ since $(-2) \mid 2$, but $a \neq b$ since $2 \neq -2$.
Therefore, the proposed divisibility property is false.



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Important
Theory

The Unique Factorization Theorem

By a German mathematician
(Carl Friedrich Gauss) in
1801.



The Unique Factorization Theorem

أي رقم اكبر من واحد إما ان يكون عدد أولي او حاصل ضرب أعداد أولية

Any integer greater than 1 either is prime or can be written as a product of prime numbers in a way that is unique except,

$$72 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 = 2 \cdot 3 \cdot 3 \cdot 2 \cdot 2 = 3 \cdot 2 \cdot 2 \cdot 3 \cdot 2$$

Theorem 4.3.5 Unique Factorization of Integers Theorem (Fundamental Theorem of Arithmetic)

Given any integer $n > 1$, there exist a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that

$$n = p_1^{e_1} p_2^{e_2} p_3^{e_3} \cdots p_k^{e_k},$$

and any other expression for n as a product of prime numbers is identical to this except, perhaps, for the order in which the factors are written.

The Standard factored Form

- **Definition**

Given any integer $n > 1$, the **standard factored form** of n is an expression of the form

$$n = p_1^{e_1} p_2^{e_2} p_3^{e_3} \cdots p_k^{e_k},$$

where k is a positive integer; $p_1, p_2, \dots, p_k$ are prime numbers; $e_1, e_2, \dots, e_k$ are positive integers; and $p_1 < p_2 < \cdots < p_k$.

Example: Write 3,300 in standard factored form.

$$\begin{aligned} 3,300 &= 100 \cdot 33 \\ &= 4 \cdot 25 \cdot 3 \cdot 11 \\ &= 2 \cdot 2 \cdot 5 \cdot 5 \cdot 3 \cdot 11 \\ &= 2^2 \cdot 3^1 \cdot 5^2 \cdot 11^1. \end{aligned}$$

Using Unique Factorization to Solve a Problem

Suppose m is an integer such that

$$8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot m = 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10$$

Does $17 \mid m$?

Solution:

Since 17 a prime in the left, it should be also in the right side.

Since we cannot produce 17 from (8,7,6,5,4,3 or 2) it should be a prime factor of m